
Code Optimization Report

HPR-LP-C Aug-20 Combined Solver

Repository	HPR-LP-C-Aug-20 dated source snapshot
Original baseline	Aug-20 archive commit 05f915038195e
Final optimized version	HPR-LP-C-SpMV0p-05f9150-presolve-combined
Report date	2026-08-20
Environment	Local source audit on macOS ARM64; CUDA execution not completed
Primary metric	End-to-end solver time, with presolve and solve time reported separately
Workflow status	Source integration complete; CUDA validation hold

This report records the exact source composition and keeps prior component evidence separate from uncollected evidence for the new combined binary. No combined runtime or correctness benefit is claimed before a matched CUDA campaign is completed.

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1 Executive summary

The Aug-20 source snapshot was found to differ from the Aug-19 baseline in exactly one production file: the automatic sparse-matrix backend selection policy. The Aug-19 presolver candidate differs from that same baseline in exactly seven GPU-presolver files. The final local candidate starts from the Aug-20 source and replaces only those seven files with their byte-exact, previously validated Aug-19 fast-presolver versions. Consequently, the Aug-20 backend policy is preserved byte-for-byte and no unrelated source is changed.

Prior evidence supports the two components separately. The Aug-19 adaptive presolver campaign matched all recorded solution-path and postsolve fields on 49 of 49 cases and reduced presolve sGM10 from 0.463171 seconds to 0.213816 seconds, a 2.1662x speedup. The supplied Aug-20 CSV reports solve-time sGM10 of 5.603869 seconds versus 6.107002 seconds in the separate Aug-19 baseline campaign, but those samples are unmatched and ten iteration counts differ. They are diagnostic component evidence only.

The source merge is complete. A CUDA build and matched combined run were not completed: the local ARM64 host has no `nvcc`, and transfer of the private candidate to the remote H100 validation host was not authorized in this task. Therefore the combined numerical verdict is inconclusive, its performance is not assessed, and the release gate is hold.

2 Baseline information

Item	Recorded value
Repository	HPR-LP-C-Aug-20 dated source snapshot
Aug-20 baseline identity	Embedded archive commit 05f915038195e9d94651eea9fcb5b6e0eba80a28; ZIP SHA-256 a872703c...91879fc
Aug-19 source identity	Git commit 2a68af144f3c37de8e22e89cb1a4662bd3bdf26a
Candidate identity	Production-tree SHA-256 a73af071...cfee8611; see the deterministic hashing command in Section 9
Baseline tree identity	Production-tree SHA-256 fc655aff...f97aeee
Dirty-state and diff identity	Dated folders are not repositories. The candidate is an isolated copy. Source patch SHA-256 is ccbe8b63...d61da4
Build configuration	Intended command <code>make GPU_SM=90 -j16</code> ; FP64 defaults retained; no build completed in this task
Compiler and dependencies	Local GNU Make 3.81; no local CUDA compiler. Prior component validation used Ubuntu 22.04, <code>nvcc</code> 11.5.119, <code>g++</code> 11.4.0, and driver 570.133.20
Hardware and software	Source audit: macOS ARM64. Prior Aug-19 component run: NVIDIA H100 PCIe SM 90. No hardware identity is claimed for the combined candidate
Dataset	Prior component evidence covers 49 Hans LP instances; the input corpus is external and was not copied into this candidate
Solver parameters	Prior policy: tolerance 10^{-6} , time limit 1000 seconds, check interval 150, GPU presolver/folding/reduced matrix enabled, automatic memory policy disabled
Benchmark policy	Required combined policy: separate diagnostics plus synchronized alternating baseline/candidate pairs on one GPU UUID. Executed repetitions: zero

The deterministic production-tree hash excludes `optimization_report/` and `.DS_Store`; it hashes sorted relative paths and their SHA-256 digests. The machine-readable record is `optimization_report/validation_status.json`.

3 Bottleneck analysis

Path	Evidence	Mechanism	Opportunity	Falsifier
GPU presolver	Aug-19 report and 98-solve campaign	Repeated whole-array transfers, serial record construction, one-thread dense-row work, redundant transpose/sort, and over-allocation	Retain packed GPU tape, order-preserving compaction, adaptive row kernels, selective workspaces, and cuSPARSE CSR-to-CSC ALG1	Any new dimension, trace, postsolve, residual, status, or iteration mismatch
Backend autotuning	Exact one-file Aug-19/Aug-20 diff and Aug-20 CSV	Deterministic specialization order can retain a slower backend; Aug-20 selects by probe time and requires a 5% advantage for fixed-degree candidates	Preserve Aug-20's measured selection policy while importing only presolver code	Candidate changes backend file hash or loses Aug-20 solve-time behavior under matched tests
Combined path	Static composition audit only	The two changes touch disjoint files but can interact through the reduced model and selected SpMV backend	Build and test their composition as a new binary	Any build failure, candidate-only failure, path change beyond the authorized policy, or runtime regression

No new profiler trace was collected for the combined binary. Amdahl-style end-to-end benefit is unavailable because the two component campaigns are not a matched composition experiment.

4 Detailed code changes

The production patch has 1,430 insertions and 816 deletions across seven files. There are no added or deleted production files.

File	Module	Change	Reason	Benefit	Validation
presolve_structs.hpp	Presolve structures	Add compact capacities and packed-record support	Avoid repeated host reconstruction and reallocation	Expected in combination	Byte-exact Aug-19 fast file
gpu_presolve.cu	Presolve driver	Selective workspaces, ordered compaction, and retained stream-safe buffers	Remove redundant allocation and serial compaction work	Expected in combination	Byte-exact Aug-19 fast file
presolve_structs.cu	Tape storage	In-place globalization and explicit capacity handling	Preserve replay ownership with fewer reallocations	Expected in combination	Byte-exact Aug-19 fast file
rule_doubleton_eq.cu	Doubleton equality	Compact tape capture and cuSPARSE CSR-to-CSC ALG1	Avoid repeated transfers, handwritten transpose, and sort	Expected in combination	Byte-exact Aug-19 fast file
rule_parallel_rows.cu	Parallel-row rule	Adaptive dense-row reduction followed by exact verification	Parallelize long-row work safely	Expected in combination	Byte-exact Aug-19 fast file
rule_primal_propagation.cu	Primal propagation	Order-preserving short/dense-row compaction and scoped statistics	Reduce serial scanning and unused allocation	Expected in combination	Byte-exact Aug-19 fast file

File	Module	Change	Reason	Benefit	Validation
<code>rule_singleton_cols.cu</code>	Singleton columns	Packed GPU records plus one bulk CPU mirror in original dual-projection order	Remove record-by-record transfer while preserving postsolve order	Expected in combination	Byte-exact Aug-19 fast file

The Aug-20 backend-selection file `src/solver/backends/autotune/backend_selection.inc.cu` has SHA-256 `5bdb28db...4e3c39` in both the baseline and candidate. All seven transplanted files compare byte-for-byte with the Aug-19 fast source. The complete production patch is `optimization_report/aug20_to_combined_source.patch`.

5 Removed or rejected changes

Item	Classification	Evidence	Reason removed or rejected
Aug-19 fast-only FOME13	Prior path regression	One reduced-active-set event changed although final metrics matched	Do not claim strict path preservation for a uniform fast binary; focused combined testing is mandatory
GPU-only singleton tape	Prior postsolve regression	<code>dlr2</code> dual residual changed	Original-order bulk CPU mirror retained
32/64 doubleton rounds	Prior algorithmic regression	Reduced dimensions and path changed	Original eight rounds retained
Persistent tape accumulator	Prior memory regression	No speedup and about 285 MB extra memory	Not present in final source
cuSPARSE CSR-to-CSC ALG2	Prior environment failure	Tested CUDA stack rejected the runtime/API path	ALG1 retained
Unrelated Aug-19 report/evidence tree	Cleanup	Directory comparison	Not copied into production source; only this combination report and exact patch were added

6 Correctness verification

6.1 Static composition checks

Check	Result	Meaning
Aug-20 baseline versus candidate production inventory	Pass	Exactly seven files differ
Aug-20 backend-selection hash	Pass	Baseline and candidate both <code>5bdb28db...4e3c39</code>
Aug-19 fast presolver content	Pass	Seven of seven files match byte-for-byte
Unexpected added/deleted production files	Pass	None
Scoped whitespace check	Pass	No whitespace diagnostics in the seven-file patch
Local CUDA build	Not run	<code>nvcc</code> is absent on the macOS ARM64 host
Remote CUDA build and smoke	Not run	Source transfer was not authorized; no source was uploaded

6.2 Solver regression verdict

Gate decision hold.

Claim and frozen policy Candidate is the seven-file source patch over the Aug-20 baseline. Required cases are the 49 Hans instances with the prior flags, plus explicit scrutiny of FOME13, `dlr2`, `degme`, and `s82`.

Evidence admission	Source identities and static composition are admitted. Prior component runs are admitted only for their original components. Combined numerical and timing evidence is incomplete.
Equivalence disposition	Observed class: undetermined . Claim disposition: unestablished . Source composition alone cannot establish E0/E1, and prior fast-only FOME13 evidence showed a trace-event difference.
Numerical verdict	inconclusive .
Performance verdict	not assessed .
Aggregation and decision rationale	No combined executable, diagnostic logs, matched baseline/candidate cases, or alternating timing pairs exist. Separate component gains cannot be multiplied or substituted for the missing composition test.
Handoff / next decisive action	Authorize source transfer to the isolated H100 validation directory, then use the remote-validation workflow for build, focused diagnostics, and the matched 49-case campaign.

7 Performance comparison

Metric/evidence	Baseline	Candidate	Speedup	Uncertainty	Disposition
Combined end-to-end solver time	Not collected	Not collected	Not available	Not available	not assessed
Combined presolve time	Not collected	Not collected	Not available	Not available	not assessed
Prior Aug-19 adaptive presolve sGM10	0.463171 s	0.213816 s	2.1662x	One sample per engine/case	Measured component result, not combined result
Prior Aug-19 adaptive total sGM10	6.940354 s	6.703507 s	1.0353x	One sample per engine/case	Measured component result, not combined result
Supplied Aug-20 solve sGM10 versus separate Aug-19 baseline	6.107002 s	5.603869 s	1.0898x	Unmatched campaigns	Diagnostic only; ten iteration mismatches
Peak memory	Not collected	Not collected	Not available	Not available	not assessed

All combined benefits remain expected until measured on one hardware/software identity with matched settings, successful exits, occupancy evidence, and a frozen alternating-pair estimator.

8 Benefits

Benefit	Evidence class	Result and basis
Runtime	Expected for combination	Preserves the byte-exact implementations responsible for the separate Aug-19 presolve and Aug-20 backend results; no composed speedup is claimed
Memory	Expected	Selective workspaces and removal of rejected persistent storage are retained; no combined peak-memory samples exist
Scalability	Qualitative	Adaptive dense-row work and packed tape structures target large presolve rows and records
Numerical stability	Unestablished for combination	Prior adaptive Aug-19 evidence passed recorded numerical gates; Aug-20's changed backend selection produced ten iteration differences versus the separate Aug-19 campaign
Maintainability	Qualitative	The production diff is isolated to seven presolver files and preserves the Aug-20 backend file verbatim
Usability	Qualitative	The candidate is a standalone Aug-20-based source directory with a complete patch and report

9 Testing and reproduction instructions

9.1 Dependencies

Use Linux x86-64 with an NVIDIA CUDA GPU, GNU Make, g++, zlib development headers, and a CUDA toolkit compatible with the target architecture. The prior H100 build used `GPU_SM=90`.

9.2 Build original and candidate

The following are exact planned commands from the Aug-20 dated folder. They were not executed in this task.

```
make -C HPR-LP-C-SpMV0p-05f9150-source GPU_SM=90 -j16
make -C HPR-LP-C-SpMV0p-05f9150-presolve-combined GPU_SM=90 -j16
```

9.3 Focused correctness diagnostics

Run baseline and candidate separately with the same physical GPU mapped to logical device 0. The first cases must include `FOME13.mps.gz`, `dlr2.mps.gz`, `degme.mps.gz`, and `s82.mps.gz`. The exact solver options are:

```
-i INPUT.mps.gz --device 0 --time-limit 1000 --tol 1e-6 \
  --check-iter 150 --max-iter 2147483647 --presolver gpu \
  --gpu-folding true --reduced-matrix true \
  --auto-memory-policy false
```

After focused diagnostics, run the frozen 49-case list as serialized baseline/candidate pairs. Diagnostics and timing must be separate. Timing requires multiple synchronized alternating pairs, identical logging, and a captured GPU UUID and competing-process snapshot.

9.4 Reproduce source identities

Run from each source root, excluding report artifacts:

```
find . -type f ! -path './optimization_report/*' \
  ! -name '.DS_Store' -print0 | LC_ALL=C sort -z | \
  xargs -0 shasum -a 256 | shasum -a 256

shasum -a 256 \
  optimization_report/aug20_to_combined_source.patch
```

9.5 Compile this report

```
latexmk -pdf -interaction=nonstopmode -halt-on-error \
  HPR_LP_C_Aug20_Combined_Optimization_Report.tex
```

10 Known limitations

- No combined CUDA binary was built; the local host has no `nvcc`.
- No combined diagnostic, complete iterate-vector comparison, memory profile, or timing pair exists.
- Remote source transfer was not authorized in this task, so no private solver source was uploaded for validation.
- The Aug-19 fast engine previously changed one **FOME13** reduced-active-set event. The Aug-19 handoff used a path-exact compatibility engine for that case; the uniform combined source has not yet been checked or adapted.
- The supplied Aug-20 CSV is a single campaign without a matching manifest in the dated folder. Its timing is diagnostic, and ten iteration counts differ from the separate Aug-19 baseline campaign.
- The 49 Hans inputs are not included. Their current hashes and availability must be recovered before a new campaign.
- No confidence interval, per-instance regression guardrail, or combined end-to-end speedup can be reported.

11 Final change summary

Item	Final record
Production files added	0
Production files modified	7; inventory in Section 4 and the source patch
Production files deleted	0
Report artifacts added	TeX report, same-stem PDF after compilation, source patch, machine-readable validation status, and captured remote-attempt evidence
Major changes	Aug-19 fast GPU-presolver implementation transplanted onto byte-exact Aug-20 source; Aug-20 backend-selection file preserved
Tests passed	Static inventory, seven byte-exact source matches, backend hash preservation, and scoped whitespace check
Tests failed or skipped	Local CUDA build unavailable; remote build, diagnostics, timing, and memory tests not run
Measured benefits	None for the combined binary; prior component measurements are reported separately
Failed targets	Combined validation and release acceptance are not yet established
Known limitations	CUDA validation hold, FOME13 compatibility risk, unmatched Aug-20 timing evidence
Tester/public packages	Not requested and not created

A Evidence index

- `optimization_report/aug20_to_combined_source.patch`: complete seven-file production patch.
- `optimization_report/validation_status.json`: machine-readable identities and verdict labels.
- `optimization_report/evidence/remote_attempt/`: local command ledger and captured access-probe artifacts. No solver source transfer or GPU run is present.

- Aug-19 sibling handoff `HPR-LP-C-Aug19-GPU3-adaptive-handoff-20260820`: prior presolver report, regression verdict, and 49-case result summaries.
- Aug-20 sibling files `HPR-LP-C-SpMV0p-05f9150-source.zip` and `05f9150_result.csv`: baseline source archive and supplied diagnostic results.